

Practice Problems

Solve the following system of linear of equation by using Gaussian Elimination:

- $x + y + 2z = 9$
1. $2x + 4y - 3z = 1$ (Solution: $x=1, y=2, z=3$)
 $3x + 6y - 5z = 0$
- 2.
- $x_2 + x_3 - 2x_4 = -3$
 $x_1 + 2x_2 - x_3 = 2$
 $2x_1 + 4x_2 + x_3 - 3x_4 = -2$ (Solution: $x_1 = -1, x_2 = 2, x_3 = 1, x_4 = 3$)
 $x_1 - 4x_2 - 7x_3 - x_4 = -19$
3. $x_1 - x_2 + 2x_3 = 4$
 $x_1 + x_3 = 6$
 $2x_1 - 3x_2 + 5x_3 = 4$ (Solution: No Solution)
 $3x_1 + 2x_2 - x_3 = 1$